



CLIMATE-AWARE MALARIA PREDICTION & EXPLAINABLE HEALTHCARE AI SYSTEM

Digital Health Africa Capstone Project Presentation



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Abstract

Malaria remains one of the most significant public health challenges across Africa, contributing substantially to morbidity and mortality despite ongoing prevention and intervention efforts. This project developed a climate-aware and spatially informed machine learning system for predicting malaria incidence across African countries using demographic, healthcare, environmental, temporal, and spatial indicators.

Three integrated datasets containing malaria, rainfall, and temperature variables were combined to create a climate-aware malaria prediction dataset consisting of 594 records across 54 African countries. Advanced feature engineering techniques including lag variables, rolling climate averages, climate interaction features, and spatial clustering were applied to improve predictive performance and capture malaria transmission dynamics.

Multiple machine learning models were evaluated using TimeSeriesSplit cross-validation, including Linear Regression, Random Forest Regressor, Gradient Boosting Regressor, and XGBoost Regressor. Among the evaluated models, the Random Forest Regressor achieved the best performance with an MAE of 30.45, RMSE of 48.16, and an R^2 Score of 0.87 on the final holdout test dataset.

SHAP (SHapley Additive exPlanations) analysis was used to improve model interpretability and identify the strongest drivers of malaria incidence prediction. The analysis revealed that previous malaria incidence, climate risk index, rainfall interactions, sanitation access, and geographic variables significantly influenced malaria transmission patterns across African countries.

A production-ready Streamlit dashboard was developed and deployed publicly to support malaria risk visualization, prediction, explainability, and policy simulation. The system demonstrates the potential of explainable healthcare AI for climate-sensitive malaria surveillance, intervention planning, and data-driven public health decision-making in Africa.

1. Introduction

Malaria remains one of the most significant public health challenges in Africa, accounting for a substantial burden of illness and mortality across the continent. According to the World Health Organization (WHO), Africa accounts for

approximately 95% of global malaria cases and deaths, with children under five years representing one of the most vulnerable populations.

The World Health Organization estimates that over 249 million malaria cases and more than 600,000 malaria-related deaths occur globally each year, with the majority concentrated in Sub-Saharan Africa. Despite ongoing interventions involving insecticide-treated mosquito nets, antimalarial treatment programs, and public health awareness campaigns, malaria continues to affect millions of individuals annually, particularly within climate-sensitive and underserved communities.

The increasing availability of healthcare, demographic, and climate-related datasets presents an opportunity to apply Artificial Intelligence (AI) and Machine Learning (ML) techniques to better understand and predict malaria transmission patterns. Predictive systems can support healthcare stakeholders by identifying high-risk regions, improving outbreak preparedness, optimizing resource allocation, and enabling evidence-based public health interventions.

This project focuses on developing a climate-aware and spatially informed Machine Learning system for predicting malaria incidence across African countries using demographic, healthcare access, climate, environmental, and temporal indicators. The study also integrates explainable AI techniques and an interactive dashboard to support malaria risk interpretation and healthcare decision-making.

By combining climate-aware modeling, temporal forecasting, spatial analysis, explainable AI, and public health intelligence, this project demonstrates the potential of AI-driven systems in supporting malaria surveillance, intervention planning, and evidence-based healthcare policy across Africa.

2. Problem Statement

Malaria continues to pose a major public health challenge across many African countries, contributing significantly to morbidity and mortality despite ongoing prevention and treatment efforts. Factors such as climate variability, limited healthcare access, poor sanitation, population dynamics, and socio-economic disparities contribute to the persistence and spread of the disease.

Traditional approaches to malaria monitoring often rely on retrospective analysis, limiting the ability to identify high-risk regions and respond proactively to outbreaks. In addition, malaria transmission patterns exhibit strong spatial and

temporal variability influenced by environmental conditions such as rainfall and temperature.

Therefore, there is a need for climate-aware and explainable machine learning systems capable of integrating healthcare, demographic, environmental, temporal, and spatial indicators to improve malaria incidence prediction and support public health decision-making across African countries.

3. Project Objective

The objective of this project is to develop a climate-aware and spatially informed Machine Learning system for predicting malaria incidence across African countries using demographic, healthcare, environmental, and temporal indicators.

4. Literature Review

Malaria remains one of the most significant public health challenges globally, with Sub-Saharan Africa accounting for approximately 95% of global malaria cases and deaths according to the World Health Organization (WHO). Climate variability, healthcare access, sanitation conditions, and population dynamics have consistently been identified as major contributors to malaria transmission patterns across African countries.

Several studies have explored the relationship between environmental conditions and malaria incidence. Climate-driven malaria forecasting research has shown that rainfall, temperature, and humidity strongly influence mosquito breeding, parasite development, and transmission intensity. Previous epidemiological studies have demonstrated that abnormal rainfall patterns and rising temperatures are associated with increased malaria outbreaks in climate-sensitive regions of Africa.

Recent advances in Artificial Intelligence (AI) and Machine Learning (ML) have enabled the development of predictive healthcare systems capable of forecasting disease burden using environmental, demographic, and healthcare datasets. Ensemble-based machine learning algorithms such as Random Forest, Gradient Boosting, and XGBoost have demonstrated strong predictive capability in epidemiological forecasting tasks involving nonlinear relationships and complex feature interactions.

Prior malaria prediction systems have primarily focused on statistical forecasting approaches, climate-only models, or regional surveillance systems with limited explainability. Several climate-health forecasting studies have integrated rainfall

and temperature variables into predictive malaria models; however, many existing systems lack explainable AI components capable of interpreting prediction outcomes for healthcare stakeholders and policymakers.

Recent healthcare AI research has increasingly emphasized the importance of explainability and transparency in predictive healthcare systems. Explainable AI methods such as SHAP (SHapley Additive exPlanations) have been widely adopted to improve interpretation of machine learning predictions, identify influential variables, and support trustworthy healthcare decision-making.

Studies involving machine learning applications in epidemiology have also demonstrated the importance of temporal forecasting and spatial analysis in infectious disease prediction. Time-series approaches using lag variables and rolling environmental averages have shown improved forecasting performance for diseases influenced by climate-sensitive transmission dynamics.

Despite these advances, gaps remain in the development of integrated climate-aware, spatially informed, and explainable healthcare AI systems specifically designed for malaria prediction across African countries. Many existing studies either lack deployment-ready healthcare dashboards, exclude explainability techniques, or fail to integrate climate, spatial, temporal, and healthcare variables within a unified machine learning framework.

This project addresses these gaps by integrating malaria, rainfall, temperature, healthcare, demographic, temporal, and spatial datasets into a unified climate-aware machine learning framework. The study further incorporates SHAP explainability, policy simulation, and a deployed Streamlit dashboard to support interpretable malaria surveillance and healthcare decision-making across Africa.

5. Methodology

This project adopted a data-driven machine learning approach to predict malaria incidence across African countries using integrated malaria, climate, demographic, healthcare, and spatial datasets.

5.1 Data Collection

This project utilized three integrated datasets: a malaria dataset containing healthcare, demographic, sanitation, and socio-economic indicators across African countries, alongside rainfall and temperature datasets capturing climate-related factors influencing malaria transmission. The datasets were merged using country

and year variables to create a unified climate-aware dataset for machine learning analysis and prediction.

5.2 Data Preprocessing

Data preprocessing was performed to prepare the integrated malaria and climate datasets for machine learning analysis. This included handling missing values, removing duplicate records, standardizing column names, correcting data types, and filtering relevant African country records. The rainfall and temperature datasets were transformed and merged with the malaria dataset using country and year variables, followed by feature scaling and encoding where necessary.

5.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to examine malaria incidence patterns, climate relationships, spatial variability, and temporal trends across African countries. Visualization techniques including histograms, line charts, scatter plots, boxplots, correlation heatmaps, geographic risk maps, and spatial cluster visualizations were used to explore feature distributions and relationships within the integrated climate-aware malaria dataset.

The distribution analysis showed that malaria incidence was highly variable across countries, ranging from near-zero values in low-burden regions to over 580 cases per 1,000 population at risk in high-burden regions. Several countries consistently exhibited malaria incidence values above 400 cases per 1,000 population at risk, highlighting substantial regional disease disparities.

Temporal trend analysis revealed a gradual decline in average malaria incidence over the study period, decreasing from approximately 207 cases per 1,000 population at risk in 2007 to approximately 176 cases per 1,000 population at risk by 2017. This represents an overall decline of nearly 15% across the observed period, although several high-burden countries remained consistently vulnerable.

Scatter plot analysis demonstrated meaningful relationships between malaria incidence and climate-related variables such as rainfall, temperature, and climate risk index, indicating the environmental sensitivity of malaria transmission. Additional analysis showed that lower sanitation access and high rural population percentages were associated with increased malaria burden.

Correlation heatmap analysis identified strong relationships between temporal lag features, climate variables, and malaria incidence, supporting the importance of climate-aware and temporal feature engineering. Geographic risk visualization further revealed spatial heterogeneity in malaria burden across African regions, while spatial clustering analysis highlighted distinct regional malaria risk patterns within the continent.

5.4 Feature Engineering

Feature engineering was performed to improve model performance and better capture the temporal, climatic, spatial, and demographic factors influencing malaria transmission across African countries.

Temporal lag features were created to represent delayed malaria and climate effects. Specifically, one-year and two-year lag variables were generated for malaria incidence, rainfall, and temperature. These lag features were created within each country grouping so that each observation only used historical values from previous years. This helped reduce the risk of temporal leakage and allowed the model to learn how previous malaria burden and prior climate conditions influenced future malaria incidence.

Rolling climate averages were also generated to capture medium-term environmental trends. A three-year rolling average was applied to rainfall and temperature variables to represent accumulated climate exposure over time rather than relying only on single-year values. This was important because malaria transmission is often influenced by preceding environmental conditions rather than isolated annual measurements.

Climate interaction features were created to capture combined environmental and population effects. These included a climate risk index, rainfall-rural population interaction, latitude-temperature interaction, and longitude-rainfall interaction. The rainfall-rural interaction was designed to reflect how rainfall effects may be amplified in rural populations where exposure, housing conditions, drainage, and healthcare access may increase vulnerability to malaria transmission.

Spatial feature engineering was used to represent geographic malaria heterogeneity across African countries. Latitude and longitude variables were retained as geographic predictors, while spatial clustering was applied to group countries with similar geographic positions. K-Means clustering was used to create spatial cluster labels based on latitude and longitude coordinates, allowing the model to capture regional malaria risk patterns beyond individual country identifiers.

Categorical country information was encoded into numerical form to make it usable for machine learning models. Country encoding was applied to represent country-level differences such as health system variation, ecological zones, surveillance capacity, and intervention history. Numerical variables were standardized using scaling techniques within the machine learning pipeline to ensure that features with different measurement units, such as rainfall, temperature, sanitation access, and population indicators, were placed on comparable scales where required.

5.5 Machine Learning Modeling

A production-ready machine learning pipeline was developed using Scikit-learn preprocessing and pipeline techniques to automate data transformation, feature scaling, encoding, and model training. TimeSeriesSplit cross-validation was used to preserve the temporal structure of the dataset and reduce data leakage during model evaluation.

Multiple regression models were trained and compared for malaria incidence prediction, including Linear Regression, Random Forest Regressor, Gradient Boosting Regressor, and XGBoost Regressor. The models were evaluated using MAE, RMSE, and R^2 Score metrics.

Among the evaluated models, the Random Forest Regressor achieved the best predictive performance and was selected as the final production-ready malaria prediction model.

5.6 Model Evaluation

Model performance was evaluated using TimeSeriesSplit cross-validation and a final holdout test dataset to preserve temporal structure and reduce data leakage. The models were assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score metrics.

Ensemble-based models outperformed Linear Regression, indicating the presence of complex nonlinear relationships within the malaria prediction task. Among the evaluated models, the Random Forest Regressor achieved the best overall performance, with an MAE of 30.45, RMSE of 48.16, and an R^2 Score of 0.87 on the holdout test dataset.

The relatively small dataset size of 594 records across 54 countries may limit model generalizability and increase susceptibility to overfitting, particularly for

ensemble-based machine learning models. TimeSeriesSplit cross-validation and temporal lag controls were implemented to reduce temporal leakage and improve model robustness.

5.6.1 Temporal Leakage Prevention and Model Robustness

Several measures were implemented to reduce the risk of temporal leakage and improve model robustness during training and evaluation.

TimeSeriesSplit cross-validation was selected instead of random K-fold validation in order to preserve the chronological ordering of observations and prevent future information from leaking into earlier training periods. This approach ensured that the model was always trained on earlier observations and evaluated on later observations, reflecting realistic forecasting conditions.

Temporal lag variables such as `malaria_incidence_lag_1`, `malaria_incidence_lag_2`, rainfall lag variables, and temperature lag variables were generated using only historical observations within each country grouping. The lag features were created using previous-year values and did not incorporate future malaria incidence information during feature generation.

The observed improvement between cross-validation performance ($R^2 \approx 0.54$) and final holdout test performance ($R^2 \approx 0.87$) may partially reflect the strong temporal persistence identified within malaria transmission dynamics, as previous malaria incidence emerged as the strongest predictor during SHAP explainability analysis. However, the relatively small dataset size and the inclusion of engineered temporal features may still increase susceptibility to overfitting in ensemble-based machine learning models.

In addition, the untuned Random Forest model outperformed the hyperparameter-tuned model on the final holdout test dataset. This may indicate that aggressive hyperparameter optimization reduced model flexibility and that the untuned model generalized more effectively within the available dataset size. The limited dataset size may also have introduced instability during hyperparameter optimization, affecting tuned model performance.

Overall, the study attempted to minimize leakage and improve temporal robustness through TimeSeriesSplit validation and controlled lag-feature generation; however, the results should still be interpreted within the limitations of the dataset size and forecasting framework.

5.7 SHAP Explainability

SHAP (SHapley Additive exPlanations) analysis was performed to improve model interpretability by identifying the variables contributing most significantly to malaria incidence prediction. The analysis showed that previous malaria incidence was the strongest predictor of future malaria burden, demonstrating strong temporal persistence in malaria transmission dynamics.

Climate-related variables, including climate risk index, rainfall interactions, and temperature lag features, also contributed meaningfully to prediction outcomes, highlighting the environmental sensitivity of malaria transmission. Additional contributions from sanitation access, drinking water access, rural population characteristics, and spatial variables reflected the influence of public health infrastructure and geographic heterogeneity across African countries.

SHAP dependence and waterfall analyses further provided interpretable explanations for both global model behavior and individual malaria predictions, strengthening the system's applicability for healthcare decision support and public health intervention planning.

5.8 Dashboard Development

An interactive Streamlit dashboard was developed and deployed to support climate-aware malaria risk visualization and healthcare decision-making across African countries.

The dashboard includes malaria trend analysis, geographic risk mapping, machine learning-based malaria incidence prediction, what-if policy simulation, SHAP explainability, and public health recommendation modules.

The deployed dashboard provides an accessible interface for exploring climate-sensitive malaria dynamics, interpreting model predictions, and supporting data-driven public health intervention planning.

6. Results and Findings

6.1 Dataset Summary

The integrated climate-aware malaria dataset consisted of:

- 594 records
- 54 African countries
- Over 30 healthcare, demographic, climate, spatial, and engineered temporal features
- Integrated malaria, rainfall, and temperature datasets

The rainfall dataset captured annual precipitation patterns across African countries, with rainfall values ranging from below 100 mm in arid regions to over 3,000 mm in high-rainfall tropical regions. The temperature dataset contained annual mean temperature measurements ranging approximately from 15°C to over 30°C across different African climatic zones.

The average malaria incidence across the dataset was approximately 188.95 cases per 1,000 population at risk, with several high-burden countries exceeding average incidence levels of 400 cases per 1,000 population at risk.

6.2 Model Performance

The following machine learning models were evaluated:

Model	Average R ² Score
Linear Regression	-9.90
Random Forest Regressor	0.54
Gradient Boosting Regressor	0.47
XGBoost Regressor	0.41

Final holdout test evaluation demonstrated superior performance for the untuned Random Forest model:

Model	MAE	RMSE	R ² Score
Untuned Random Forest	30.45	48.16	0.87
Tuned Random Forest	57.36	70.54	0.73

The untuned Random Forest model was selected as the final production-ready malaria prediction model.

6.3 SHAP Explainability Findings

SHAP analysis revealed that the strongest predictors of malaria incidence included:

- Previous malaria incidence (temporal persistence)
- Climate risk index
- Rainfall-rural interaction
- Geographic location variables
- Sanitation access
- Temperature lag variables
- Drinking water access

The SHAP summary analysis demonstrated that previous malaria incidence was the most influential predictor, contributing over 200 SHAP units in individual prediction explanations. Climate-related variables, including climate risk index and rainfall-rural interaction features, consistently contributed positive SHAP values toward increased malaria risk predictions.

The SHAP waterfall explanation showed that previous malaria incidence contributed approximately +209.69 SHAP units toward one high-risk malaria prediction, while climate risk index, rainfall-rural interaction, sanitation access, and longitude variables also contributed positively to elevated malaria burden predictions.

Top SHAP feature importance rankings further demonstrated that temporal persistence, climate-sensitive variables, sanitation access, and spatial features were among the strongest drivers influencing malaria transmission dynamics across African countries.

The SHAP dependence analysis demonstrated that higher climate risk values produced increasingly positive contributions toward malaria incidence prediction, supporting the climate-sensitive nature of malaria transmission.

The waterfall explanation further showed that previous malaria incidence contributed strongly toward elevated malaria predictions, demonstrating significant temporal persistence in malaria burden across African countries.

6.4 What-If Policy Simulation

A what-if policy simulation was conducted to evaluate the sensitivity of malaria incidence predictions to sanitation-related interventions. The simulation increased the percentage of the population with access to basic sanitation services for selected country-year observations and generated updated malaria incidence predictions using the trained Random Forest model.

One deployed dashboard scenario utilized Mozambique (2008) as a sample intervention case. The baseline predicted malaria incidence was approximately 399.79 cases per 1,000 population at risk. After simulating a 20% increase in sanitation access, the predicted malaria incidence decreased to approximately 399.19 cases per 1,000 population at risk, representing an estimated reduction of 0.60 cases per 1,000 population at risk.

Additional simulation scenarios produced similar reductions ranging approximately from 0.60–0.71 cases per 1,000 population at risk. Although the reduction magnitude was relatively modest, the simulation demonstrated directional sensitivity of the model to sanitation-related interventions rather than causal impact estimation.

These findings suggest that sanitation improvements may contribute to lowering malaria burden when considered alongside other influential factors such as climate variability, historical malaria persistence, healthcare access, and geographic conditions. The policy simulation therefore serves primarily as a decision-support and scenario-analysis tool rather than a definitive causal intervention model.

7. Policy Recommendations

Based on the findings from the climate-aware malaria prediction system, several public health recommendations are proposed to support malaria prevention and intervention planning across African countries.

- Strengthen malaria surveillance in historically high-burden regions to improve early detection and outbreak preparedness.
- Integrate rainfall, temperature, and climate-risk monitoring into malaria early-warning systems.
- Improve sanitation and drinking water access in vulnerable communities to reduce environmental and infrastructure-related malaria risk.
- Prioritize rural and climate-sensitive regions for targeted intervention programs and healthcare resource allocation.
- Utilize geographic risk mapping and predictive analytics to support data-driven public health decision-making.
- Apply explainable AI techniques to improve transparency and interpretation of healthcare prediction systems for policymakers and healthcare stakeholders.

The what-if policy simulation demonstrated that sanitation improvements alone could reduce predicted malaria incidence by approximately 0.60–0.71 cases per 1,000 population at risk in selected intervention scenarios. Combined climate-sensitive interventions involving sanitation improvement, enhanced surveillance, and targeted rural healthcare support may contribute to broader malaria burden reduction across high-risk regions.

The deployed healthcare AI dashboard can further support policymakers by enabling geographic malaria risk visualization, climate-aware outbreak monitoring, and explainable AI-driven intervention planning for resource allocation and malaria prevention strategies.

8. Ethical Considerations

Artificial Intelligence systems used in healthcare and public health forecasting require careful consideration of transparency, fairness, interpretability, and responsible deployment. This project recognizes several ethical considerations associated with climate-aware malaria prediction and healthcare AI deployment across African countries.

First, the model predictions generated in this study are non-causal and should not be interpreted as definitive epidemiological outcomes or clinical decisions. The system is intended to support malaria surveillance and public health planning rather than replace healthcare professionals, epidemiologists, or national malaria control programs.

Second, limitations within the underlying datasets may introduce bias into prediction outcomes. Variations in malaria surveillance quality, healthcare infrastructure, testing availability, and reporting completeness across African countries may affect the accuracy and representativeness of the training data. Consequently, model predictions may not fully reflect true malaria burden in regions with underreporting or limited healthcare access.

Third, explainable AI techniques such as SHAP were incorporated to improve model transparency and support responsible healthcare interpretation. By identifying the variables contributing most strongly to malaria predictions, the system enables healthcare stakeholders and policymakers to better understand the factors influencing model outputs.

In addition, the deployed Streamlit dashboard was designed primarily as a decision-support and exploratory analytics tool rather than an automated intervention system. Public health recommendations generated from the model should therefore be interpreted within broader epidemiological, environmental, and healthcare contexts.

Finally, responsible deployment of healthcare AI systems requires ongoing validation, monitoring, and contextual interpretation by healthcare professionals and public health stakeholders to ensure safe, fair, and appropriate use within real-world malaria surveillance and intervention planning environments.

9. Limitations

Several limitations were identified during the study:

- The relatively small dataset size may limit model generalizability and increase susceptibility to overfitting, particularly for ensemble-based machine learning models.
- Country-level aggregation may not fully capture local malaria transmission dynamics.
- The model relies on reported malaria incidence, which may underestimate actual disease burden in areas with limited testing infrastructure.
- Additional environmental variables such as humidity and vegetation indices were not included due to data availability constraints.
- Streamlit deployment environment differences introduced minor reproducibility considerations during online deployment.

Despite these limitations, the model demonstrated strong predictive capability and meaningful public health interpretability

10. Live Dashboard Deployment

A production-ready Streamlit dashboard was successfully deployed publicly to demonstrate the climate-aware malaria prediction system and support interactive healthcare decision-making.

Dashboard Features:

- Dashboard Overview
- Geographic Malaria Risk Mapping
- Malaria Incidence Prediction Tool
- What-If Policy Simulation
- SHAP Explainability
- Public Health Policy Recommendations

The deployed What-If Policy Simulation module allows users to select individual African countries and years, simulate sanitation-access improvements, and estimate resulting changes in malaria incidence predictions.

The deployed dashboard demonstrates how machine learning, climate intelligence, explainable AI, and public health analytics can be integrated into a real-time healthcare decision-support system.

Public App Link: <https://malaria-app-app-hp7r78ub8agnjku4n9stpf.streamlit.app/>

GitHub Repository: <https://github.com/drsam-israel/malaria-streamlit-app>

Potential Future Work

Several opportunities exist for extending and improving the current climate-aware malaria prediction system in future studies.

- Integrate additional environmental variables such as humidity, vegetation indices, land surface temperature, satellite imagery, and mosquito habitat indicators to strengthen climate-sensitive malaria forecasting.

- Explore advanced deep learning and spatiotemporal forecasting approaches such as LSTM networks, Temporal Fusion Transformers, Graph Neural Networks (GNNs), and hybrid climate-health forecasting models.
- Incorporate real-time climate APIs and automated data pipelines to support continuous malaria surveillance and live forecasting capabilities.
- Improve causal inference and intervention modeling by integrating epidemiological simulation methods and public health intervention datasets.
- Extend the dashboard with geospatial drill-down capabilities, regional forecasting modules, and automated outbreak-alert systems for healthcare stakeholders and national malaria control programs.
- Integrate Large Language Models (LLMs) and conversational AI interfaces to support interactive public health analytics and healthcare stakeholder interpretation.
- Explore mobile and low-resource deployment strategies to improve accessibility of climate-aware healthcare AI systems within underserved regions across Africa.

11. Conclusion

This project developed a climate-aware and spatially informed machine learning system for predicting malaria incidence across African countries using demographic, healthcare, environmental, temporal, and spatial indicators. The study demonstrated that malaria transmission is strongly influenced by historical malaria burden, climate variability, sanitation access, and geographic factors.

Among the evaluated models, the Random Forest Regressor achieved the best predictive performance and was selected as the final production-ready model. The final model achieved an R^2 Score of 0.87, indicating strong predictive capability in forecasting malaria incidence across African countries. The integration of SHAP explainability improved model transparency by identifying the key drivers influencing malaria predictions, while the deployed Streamlit dashboard provided an interactive platform for malaria risk visualization, prediction, and public health decision support.

Overall, the project highlights the potential of explainable healthcare AI systems in supporting climate-sensitive malaria surveillance, intervention planning, and data-driven public health decision-making across Africa.

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